S.No: 8	Exp. Name: ***Matrix Multiplication***	Date:2026-07-10

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Aim:

Write a program to multiply two matrices A and B. The program should read their dimensions and elements, check whether matrix multiplication is possible, and display the resulting product matrix.

If the number of columns in matrix A is not equal to the number of rows in matrix B, the program should skip reading the elements of the second matrix and print "Invalid input".

Input Format:

- The first line contains two integers, m and n, representing the dimensions of matrix A.
- The next m lines each contain n space-separated integers, representing matrix A.
- The next line contains two integers, n and p, representing the dimensions of matrix B.
- The next n lines each contain p space-separated integers, representing matrix B.

Output Format:

- Print the resulting matrix after multiplication, where each row is displayed on a new line with space-separated integers.
- If multiplication is not possible (number of columns of A ≠ number of rows of B), print "Invalid input".

Source Code:

CTC77533.c

```
#include <stdio.h>

#define MAX 100

int main() {
    int m1, n1;
    if (scanf("%d %d", &m1, &n1) != 2) return 0;

    int A[MAX][MAX];
    for (int i = 0; i < m1; i++) {
        for (int j = 0; j < n1; j++) {
            if (scanf("%d", &A[i][j]) != 1) return 0;
        }
    }

    int m2, n2;
    if (scanf("%d %d", &m2, &n2) != 2) return 0;

    if (n1 != m2) {
        printf("Invalid input\n");
        return 0;
    }

    int B[MAX][MAX];
    for (int i = 0; i < m2; i++) {
        for (int j = 0; j < n2; j++) {
            if (scanf("%d", &B[i][j]) != 1) return 0;
        }
    }
```

```
    int C[MAX][MAX] = {0};

    for (int i = 0; i < m1; i++) {
        for (int j = 0; j < n2; j++) {
            for (int k = 0; k < n1; k++) {
                C[i][j] += A[i][k] * B[k][j];
            }
        }
    }

    // Explicit space after every number layout configuration
    for (int i = 0; i < m1; i++) {
        for (int j = 0; j < n2; j++) {
            printf("%d ", C[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
2 3
1 2 3
4 5 6
3 2
7 8
9 10
11 12
58 64
139 154

Test Case - 2
User Output
1 2
12 45
1 2
Invalid input